

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-1374
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	ACL script uses uniform 4-space indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Configuration grouped by module: Users, Roles, Tables, ACLs.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Roles (bb1–bb4) map to Read/Create/Write/Delete; table name u_institution_details is descriptive.
4	Function / Method Names	Functions are descriptively named	Partial	ACL script is a single self-invoking function; built-in gs.hasRole() is used.
5	Code Comments	Inline and block comments explain logic	Yes	Each condition in the ACL script has a comment.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate ACL for each operation (Read, Create, Write, Delete).
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate ACLs or unused roles.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Script defaults to return false (deny) for all other users.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Read ACL script (role check with gs.hasRole)	JavaScript (ServiceNow)	Any other table needing role-based read restriction	High
2	Roles bb1–bb4	ServiceNow Roles	Assignable to other users and ACLs	High
3	Create / Write / Delete ACL pattern	ServiceNow ACL	Other custom tables (change the Name and role)	Medium
4	Data condition (Branch is EEE)	ServiceNow ACL condition	Other field-value based restrictions	Medium
5	Institution Details table (u_institution_details)	ServiceNow Table	Other student / institution based modules	Low

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Well-organized modules and consistent indentation.
Readability	4	Clear names and commented script.
Reusability	4	Script and ACL pattern can be reused on other tables.
Documentation / Comments	4	Inline comments and step-by-step setup documented.
Overall Score	4 / 5	Good quality, maintainable solution.